

Solvent-based plastic recycling technologies

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Solvent-based recycling approaches are receiving industrial and academic interest for their ability to produce high-quality plastic resins from a variety of plastic waste sources without breaking the polymer chains. Here we highlight the development of solvent-based technologies, focusing on the underlying principles, techno-economic and life-cycle analyses, and commercialization. The basic steps in solvent-based recycling include plastic size reduction, plastic dissolution, filtration or centrifugation, and optional additional cleaning steps such as adsorption, precipitation and solvent removal. Impurities that build up in the solvent must also be removed. The goal of solvent-based technologies is to produce a high-quality resin without plastic contaminants or other added substances. Disadvantages of these solvent-based technologies are their physicochemical complexity and the difficulty in scaling up to achieve continuous operations with high polymer and solvent yields and throughputs. Chemical engineering is thus critical in bringing solvent-based recycling technologies to market.

Global annual plastic production reached 413.8 million metric tons in 2023, while the recycling rate was only 9.6% (ref. 1). Globally, most plastic waste is either landfilled or disposed of in open pit dumps. It is estimated that 2–5% of all plastic produced ends up in the ocean². Mechanical recycling is the main current approach to recycle plastics³. In mechanical recycling, plastics undergo sorting, separation, grinding, cleaning by (friction) washing, sink–float separation (to separate heavier polymers such as polyethylene terephthalate (PET) from polyolefins), drying and then re-extrusion. Although some color sorting is possible before mechanical recycling, the recycling process does not remove colors or other common additives, nor is it capable of eliminating many other impurities such as adhesives. Beyond mechanical recycling, several chemical recycling technologies, such as plastic pyrolysis and polyester methanolysis^{4,5}, are being developed and commercialized. These technologies are typically integrated with existing petrochemical infrastructure, such as steam-cracking or PET polymerization⁶. Researchers have discussed various plastic

waste recycling technologies and their implications. For instance, Tiwari et al. provided a comprehensive analysis of mechanical and chemical recycling methods of plastic waste, highlighting the challenges in plastic sorting and segregation⁷; Zhang et al. covered the technological mechanisms and development status of mechanical and chemical recycling technologies⁸; additionally, Li et al. provided a comprehensive review of plastic recycling, detailing the chemistry, engineering and technological aspects of various plastic recycling methods⁹.

Flexible packaging plastic is a complicated waste stream to recycle because moisture, dirt, other polymer layers, adhesives and inks build up during the mechanical recycling process^{10,11}. For example, multilayer flexible packaging is composed of multiple polymer and adhesive layers, each with varying chemical compositions and properties. These complex combinations can make it challenging to recycle films and to produce high-quality products via mechanical or chemical recycling methods¹². Solvent-based recycling, also called dissolution-based recycling, is a promising solution that separates individual polymers

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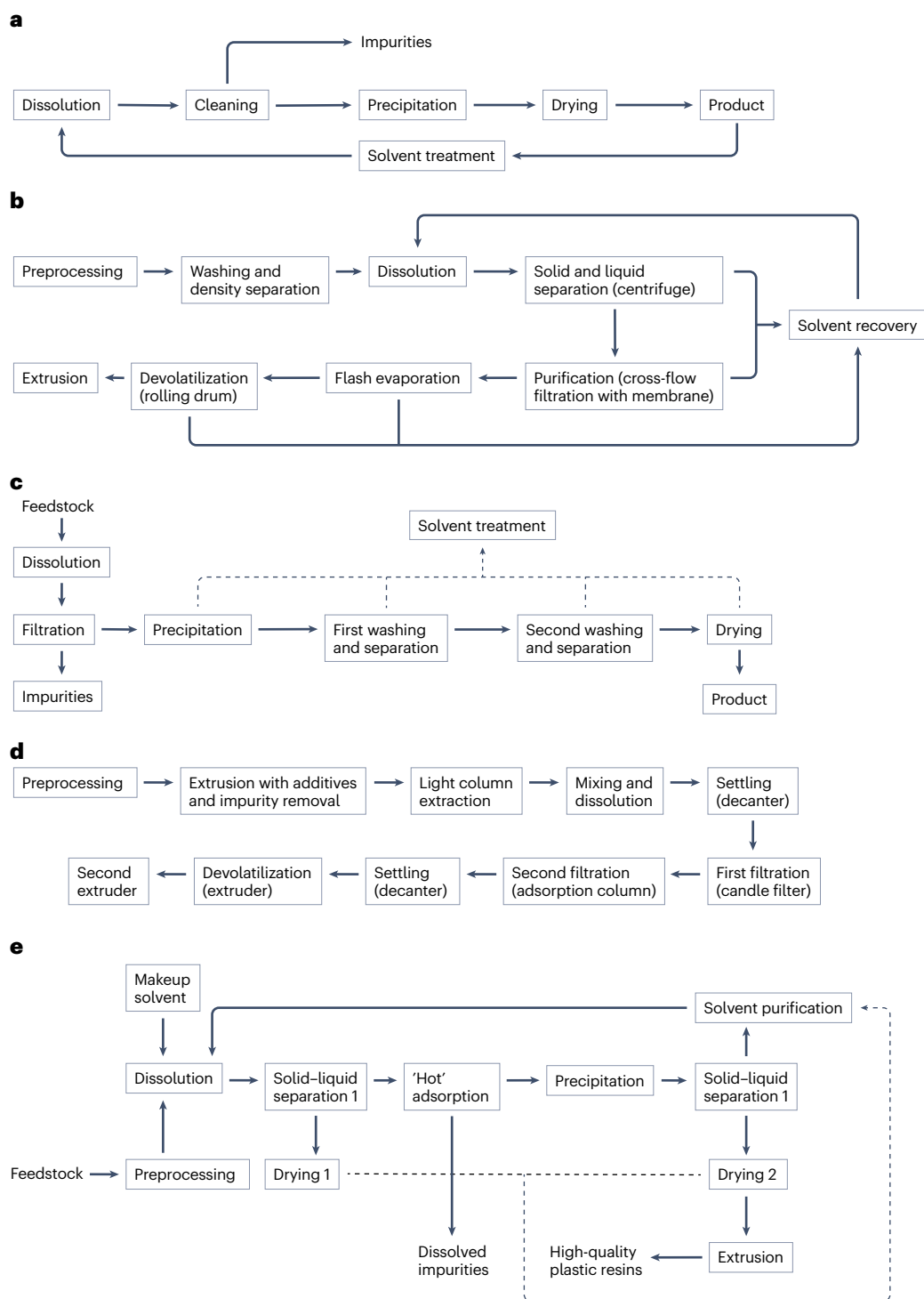


Fig. 1 | Simplified block flow diagrams of several current commercial dissolution-based plastic recycling plants. a, CreaSolv¹²⁶. **b**, APK Newcycling¹²¹. **c**, Polystyvert⁷³. **d**, PureCycle⁷⁷. **e**, STRAP³⁰. Specific processes will use unique equipment and may skip, add to or combine steps shown above.

from multipolymer streams and simultaneously allows deep cleaning of the dissolved polymers. In addition to films, this approach is well suited to other combinations of materials such as mixed plastic bales¹³, electronic^{14,15} and automotive waste¹⁶, and multicomponent flooring¹⁷. Dissolution-based recycling can also be applied to monomaterial waste streams to perform deeper removal of impurities, while keeping the polymer chains intact.

Dissolution-based plastic recycling has several advantages compared with mechanical recycling. The process produces a high-quality

virgin-like resin where impurities, other polymers and other unintentionally added substances (such as color bodies, often referred to as ‘non-intentionally added substances’) can be removed. Although dissolution-based recycling is often still classified under ‘mechanical’ recycling because it does not break the polymer chains, traditional mechanical recycling cannot achieve the same level of impurity removal. Therefore, dissolution-based recycling is categorized as ‘physical’ recycling¹⁸. This approach recovers high-quality polymers, leaving polymer chains intact. A range of thermoplastic resins has been

Table 1 | Recent experimental studies of dissolution-based recycling for different materials

Feedstock	Target material	Solvent	Temperature (°C)	Notes	Refs.
Fiber-reinforced composites or electronic packaging	Anhydride-epoxy	1,5,7-Triazabicyclo[4.4.0]dec-5-ene alcohol	150–180 °C	Dissolution of epoxy thermoset was achieved by selective bond cleavage via transesterification.	103
Multilayer flexible packaging	LDPE, ethylene-vinyl acetate (EVA)	<i>N,N</i> -Dimethylcyclohexylamine	50–90 °C	LDPE and Al are recovered from food packaging with switchable hydrophilicity solvents and ultrasonic treatment.	104
Electronic waste	Polycarbonate, polyamide (PA)	<i>N</i> -Methyl pyrrolidone (NMP), dichloromethane	38–80 °C	PA was recovered as the non-dissolved fraction. Methanol, an antisolvent, was added to the solvent to precipitate the polycarbonate.	15
Rigid and foamed materials	PS	Ethyl acetate	77 °C	PS recovery: (1) solvent evaporation by contact with water in a tubular reactor and (2) solvent evaporation during extrusion.	105
Carbon-fiber-reinforced plastics	PA6, polyurethane, epoxy matrix	CreaSolv formulations	140–280 °C	Different proprietary formulations dissolved PA6, polyurethane or epoxy matrix and the carbon-fiber-reinforced plastics were recovered as the non-dissolved fractions.	20
Automotive materials	Polybutylene terephthalate	CreaSolv formulations	190 °C	Polybutylene terephthalate was selectively dissolved, and Cu was recovered as the non-dissolved fraction.	16
Disposable containers	Expanded polystyrene (EPS)	Essential oils	RT	Methanol was added as an antisolvent to recover the dissolved PS.	36
Plastic scrap	PS adhesives	Ethyl acetate	23 °C	After dissolution of PS-based adhesives, polymers such as HDPE, LDPE and PP were separated by density.	25
Waste electrical and electronic equipment	Brominated flame retardants	Isopropanol, toluene	132–153 °C	Removal of tetrabromobisphenol A in different waste electrical and electronic equipment via dissolution.	23
Waste electrical and electronic equipment	Brominated flame retardants	1-Propanol and heptane mixture, 4:1 (vol/vol)	110–130 °C	Removal of tetrabromobisphenol A present in acrylonitrile butadiene styrene. Conditions were optimized by looking at microwave- and ultrasound-assisted extractions.	22
Aeronautic materials	Poly(lactide (PLA)	Tetrahydrofuran (THF)	66 °C	PLA was separated from PLA/lunar regolith simulant composites by selective dissolution in THF.	106
Polymer resins	Plastic additives: antioxidants, plasticizers, flame retardants	Various: dichloromethane, THF, chloroform, acetonitrile, toluene, depending on the additive	RT–170 °C	The study summarizes the extraction of various plastic additives by selective dissolution.	85
Pharmaceutical materials	PVC	Lactic acid, lactic acid-choline chloride	80–120 °C	Separation of plastic components and aluminum from pharmaceutical materials.	107
Multilayer packaging films	PE, EVOH, PET	Toluene, dimethylsulfoxide (DMSO)	95–110 °C	Recovery of polymers from rigid multilayer films. Solvent selection from computational modeling.	108
Plastic bottles	PET	Gamma-valerolactone (GVL)	120–170 °C	Recycling of PET by swelling and dissolution. Water was added as an antisolvent to precipitate the polymer.	80
Electronic waste	PS, acrylonitrile butadiene styrene	Dichloromethane	RT	Methanol and ethylene glycol were used as antisolvents for polymer precipitation.	14
Waste printed circuit boards	Brominated epoxy resin	NMP, DMSO, dimethylformamide (DMF), dimethylacetamide (DMAC)	200 °C	Delamination of multilayer waste printed circuit boards to separate fiberglass and copper layers. The size of waste printed circuit boards and conditions were optimized, along with microwave treatment.	109
Multilayer packaging with paper	LDPE	Xylene	85 °C	Recycling of Tetra Pak packaging, which contains LDPE, Al and paperboard. The paper is recycled via a hydro-pulping process and the LDPE and Al are recovered after the selective dissolution of LDPE. The LDPE is precipitated with the addition of <i>i</i> -propanol.	110
Multilayer packaging films	Polyurethane adhesives	Formic acid	50–75 °C	Delamination of multilayer packaging with carboxylic acids, recovering Al, PET, PP, PE from different samples after the separation.	111

Table 1 (continued) | Recent experimental studies of dissolution-based recycling for different materials

Feedstock	Target material	Solvent	Temperature (°C)	Notes	Refs.
Post-consumer carton packaging	PE	Xylene	85 °C	Post-consumer packaging consisted of paper, PE and PA. The paper fraction was converted to levoglucosenone and furfural in THF/H ₂ SO ₄ and the PE was recovered after dissolution in xylene. The non-dissolved solids, mainly PA, were converted to caprolactam by pyrolysis.	112
Disposable facemasks	PP	Toluene	110 °C	DMAC is used to remove the color in the PP after dissolution and precipitation.	91
Pure resins and plastic waste containers	LDPE, PP	D-Limonene, α -pinene	130 °C	The polyolefins were dissolved in green solvents, and these were then utilized for membrane fabrication.	101
Plastic bottles	PET	<i>o</i> -Chlorophenol, <i>m</i> -cresol, phenol, trifluoroacetic acid	25–173 °C	A MATLAB program based on an iterative algorithm was developed to guide solvent selection with solubility predictions. The program was named Hansen solubility parameters temperature predictive tool (HSPs-TPT).	113
Foams	EPS	Thinner and acetone	RT	The dissolution of EPS foam board was dissolved in a thinner:acetone mixture in a 3:1 volume ratio.	114
Plastic packaging	HDPE	Limonene	140 °C	Removal of blue and orange colorants from plastic packages by dissolving HDPE in limonene. Tri-alcohols were found to be optimal antisolvents for HDPE precipitation and pigment separation.	24
Laminated glass components	Polyvinyl(butylal) (PVB)	2-Propanol and ethyl acetate	45 °C	Demonstrated the dissolution of PVB in binary mixtures of 2-propanol and ethyl acetate. PVB solutions could have applications in glues for different materials or paint additives.	115
Textile waste	Elastane	Tetrahydrofurfuryl alcohol	100 °C	In the presence of other textile materials such as polyesters, cotton or wool, other solvents were considered. These included NMP, GVL, DMF, DMAC and Cyrene.	92
Printed multilayer films	PE, EVOH, PET	Dodecane, DMSO, water, GVL	95–125 °C	Recovery of polymers from flexible printed multilayer films. GVL is used to remove polyurethane-based inks from the PET component.	83
Packaging, household and automotive applications	PS, high-impact polystyrene (HIPS)	Xylene, D-limonene	RT	Filtration and centrifugation processes were included as cleaning techniques for contaminants that did not dissolve.	116
Plastic films	Polymer resins in ink binders: PVB, PVA, polyurethane, among others	Various: formic acid, sulfuric acid, cetyltrimethylammonium bromide solution, ethyl acetate, acetone, among others	RT	Comprehensive study on the dissolution of common polymer resins used as binders in inks and their removal from plastic substrates.	21

recycled with this approach, from flexible and rigid waste streams, including polyethylene (PE), polypropylene (PP), ethylene vinyl alcohol (EVOH), PET, polycarbonate, polystyrene (PS), polyvinyl chloride (PVC) and polyurethane^{19,20}. However, dissolution-based technologies also have several challenges that must be overcome for them to be implemented commercially.

In this Review, we discuss current academic research, industrial processes and future research needs of dissolution-based recycling technologies. We first review the basic underlying principles and commercial activities of dissolution-based technologies. We then discuss thermodynamic principles, experimental laboratory studies, and techno-economic analysis (TEA) and life-cycle assessment (LCA) studies. We close with key recommendations for future research.

Process and unit operations overview

Figure 1 shows simplified block flow diagrams of current commercial dissolution-based plastic recycling plants. There are variations in the

technology, but in general dissolution-based recycling of waste polymers involves several fundamental steps: (1) preprocessing (such as shredding/downsizing, densification, metal removal, washing and drying), (2) dissolution of targeted polymer, (3) separation of the dissolved polymer from the remaining non-dissolved material, (4) precipitation of the dissolved polymer, (5) separation of the precipitated polymer from the solvent, (6) drying and solvent purification from the non-dissolved polymer and precipitated polymer and (7) resin production. This section discusses each step of the process, highlighting some of the key technical considerations, state-of-the-art approaches and remaining challenges. More details of the processes are covered in the following sections.

Feedstocks and preprocessing

Dissolution-based recycling methods have been reported in the literature for a wide range of feedstocks. Table 1 references recent literature reflecting some of the most widely recognized work on this topic with

feedstocks including electronic waste, plastic packaging, foams, plastic bottles, automotive materials and others. Additionally, solvents have been used for the removal of colorants, adhesives, plastic additives and odor^{21–25}. While the technology needs time to mature and expand, dissolution-based recycling will likely fill a role in processing complex but low-variability waste streams, such as many of those listed in Table 1, that allow simplification of the preprocessing steps.

Preprocessing of plastic waste involves the following steps: (1) collection and transportation, (2) sorting and separation, (3) size reduction (typically to less than 2 mm)²⁶, (4) cleaning, (5) drying and (6) additional processing (such as decontamination, densification, microperforation and thermal conditioning). Quality control is critical to ensure that the undesired components are removed. The collection and sorting of the waste feedstock is one of the most challenging steps in this process. It plays a pivotal role in the whole recycling process to determine the solvent type and operating conditions. Most material recovery facilities use a series of sorting processes including optical methods such as near-infrared spectroscopy, which typically have accuracies of above 90% at object level (an object can consist of different polymers)²⁷. Dissolution-based recycling plants can process sorted bales, and even shredded and washed waste, or undertake some of these steps in house⁹.

Dissolution

The shredded plastics are mixed with a potentially heated solvent in a dissolution vessel. Solvent choice and temperature depend on the target polymer. Ideally, only the targeted polymer is dissolved, without dissolving other polymers and small additives including pigments, which facilitates their separation. The viscosity of the solution, an important parameter for the whole process, can be influenced by concentration, temperature, pressure and polymer–solvent interactions²⁸. Generally, for dissolution-based plastic recycling, the recommended polymer concentration is 5–20 wt% (ref. 28). A concentration above 10 wt% often results in a very viscous solution, depending on the polymer type, molecular weight and operating conditions. For example, at 155 °C and 20 MPa, the viscosity of the PE solution (1 wt%, weight-averaged molecular mass of 2,150 g mol⁻¹) is approximately 0.125 mPa s, and it increases to 0.15 mPa s when the pressure is raised to 40 MPa (ref. 28); for the combination of PS/*o*-xylene at room temperature (RT), the entanglement concentration is 13.9 wt%, and the viscosity is noticeably increased at this concentration, reaching ~0.9 Pa s (ref. 29). High viscosity of the polymer solution often leads to difficulties in material flow, separation and post-process cleaning. Polymer solutions are typically categorized as either Newtonian or non-Newtonian fluids on the basis of how the fluid viscosity changes upon the application of shear stress. Researchers have proposed the Carreau and Yasuda–Armstrong–Cohen model to describe both Newtonian and non-Newtonian fluids²⁸. These model parameters require modification for each polymer–solvent combination, as the complexity of the polymer solution viscosity requires further study²⁸.

Separation of dissolved polymer from undissolved material

The undissolved material needs to be separated from the viscous polymer solution. Equipment such as centrifuges, strainers, centrifugal sifters and candle filters can be used for this step. For example, centrifugal filters with a rotary knife/brush with fixed screens can be used in this step. In such equipment, material is fed into the chamber, where the rotating knife/brush pushes the material against the screen. Solvent and finer particles pass through the screen to the next step, while larger undissolved particles are pushed out³⁰. As an alternative to centrifugation, filtration is often applied to remove undissolved particles. Filter media must be carefully selected to avoid clogging, often consisting of multiple steps aimed at removing increasingly smaller particles. A common filtration method is candle filters, which use cartridges to filter fluid and trap impurities and particles. These candle filters

can often remove smaller-sized impurities (down to a nominal size of 0.5 µm) from the fluid³¹.

Post-cleaning dissolved substances

Additional cleaning steps may be necessary to remove contaminants that are not removed by filtration. Such contaminants often include dissolved dyes, organic and inorganic pigments, residue pigments and other undissolved particles that have passed through the previous filter size. Currently, adsorption is the predominant method to achieve this step. For example, Procter and Gamble has reported using adsorbents selected from metal oxides, covalent organic frameworks, metal–organic frameworks and zeolitic imidazolate frameworks to remove contaminants from the polymer solution³². Other adsorbents that can be used include activated carbon, Fuller's earth and/or quartz sand to remove the colorants and other dissolved additives from the polymer solution³³. Another approach for post-cleaning is ion exchange; however, it is not yet widely adopted, with rather limited information available³⁴.

Precipitation of dissolved polymer and solvent separation

The cleaned polymer is then precipitated from the solution by one of three different approaches: (1) adding antisolvent, (2) changing the temperature or pressure of the solvent or (3) evaporation of the solvent. The selection of the appropriate method in this step depends on the types of polymer and solvent used in the process. If possible, antisolvents should be avoided because they introduce the need for costly separation steps, such as distillation, to separate solvent from antisolvent. The second method is useful if the polymer–solvent interaction is highly dependent on temperature or pressure, and the third option is often preferred when the solvent has a low boiling point. More information on the methods used for various polymer–solvent systems is included in Tables 1 and 2. During this stage, a notable challenge is the gelation of polymers (a process where the precipitated polymer and solvent form a single gel-like phase instead of distinct liquid and solid phases) during precipitation, which leads to complications in subsequent separation of precipitates from the solvent³⁵. After precipitation, the precipitated polymer will be separated from the solvent. Researchers have used methods such as decantation³² and filtration to remove the solvent³⁶. Even in complex mixed plastic wastes that contain ten different plastics, laboratory-scale experiments recovered 89 wt% or higher for each polymer³⁷.

Solvent recovery and purification

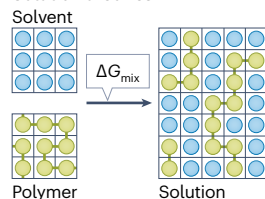
After the separation steps, the undissolved material and precipitated polymer contain certain amounts of solvent. These remaining solvents must be recovered for safety, environmental and economic reasons. Solvents are removed from the solid materials by further drying, often with a vacuum followed by condensing the vapors; however, quantitative studies on this aspect are still lacking. In addition, impurities in the solvent after dissolution can include organic or very fine inorganic color bodies, phthalates, rubbers/gels and smaller fragments of the plastics. Low levels of impurities (less than 0.5 wt%) can be removed by adsorption³². The solvent often needs to be purified before reuse to remove the impurities or any antisolvent used in the process. In this case, a distillation step is required to separate the solvent from the antisolvent. In the case of precipitation without antisolvent, the solvent might be recovered by a simpler evaporation step. Although solvent purification technologies have undergone extensive studies, integration of these methodologies into recycling processes presents challenges due to the diverse spectrum of contaminants produced during the process.

Final resin properties

The final purified and extruded resins need to be qualified in the downstream application to make sure the products meet consumer needs. The quality (for example, purity and resulting mechanical and thermal

Table 2 | Summary of the existing dissolution-based plastic recycling pilot-scale and industrial plants

Technology	Nameplate capacity and location	Feedstock	Product	Solvent and dissolution condition	Refs.
CreaSolv	Unilever: 1,000 ton yr ⁻¹ in Indonesia ¹¹⁷	Sachet packaging waste (primarily PE)	PE	Not disclosed, but selected from the group comprising ketones, ether, cycloalkanes and esters	66
	PolyStyreneLoop Cooperative: 3,000 ton yr ⁻¹ in the Netherlands ¹¹⁸	PS foam waste	PS	Not disclosed, but selected from the group comprising ketones, ether, cycloalkanes and esters	66
	Circular Flooring: 120–160 ton yr ⁻¹ in Germany ¹¹⁹	Post-consumer PVC floor covering	PVC	Not disclosed, but selected from the group comprising ketones, ether, cycloalkanes and esters	66
PureCycle	49,000 ton yr ⁻¹ starting operation in the USA and 2×59,000-ton plants under construction in the USA ¹²⁰	Industrial PP waste; carpet and textile waste	PP	Supercritical butane at 110–170 °C and 7.58–14.48 MPa; supercritical propane at 80–220 °C and 13.79–55.16 MPa	32
APK Newcycling	8,000 ton yr ⁻¹ in Germany	Mixed plastic wastes	LDPE and PA	For LDPE: <i>n</i> -heptane at 90 °C For PA: propylene glycol at 140 °C	70,121
Polystyvert	600 ton yr ⁻¹ in Canada	Post-consumer and post-industrial PS waste	PS	<i>p</i> -Cymene at 98–110 °C	73,75
STRAP	200 ton yr ⁻¹ in the USA	Post-consumer mixed plastic wastes	LDPE and PP	For LDPE: xylene at 85 °C For PP: xylene at 115 °C	9,30

Fundamental theories**Solution theories****Solubility parameters**

+ Polymer • Solvent Soluble region

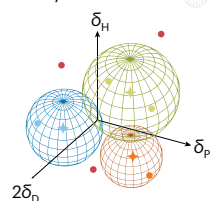
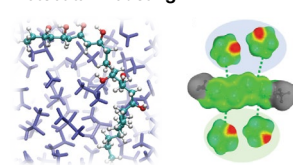
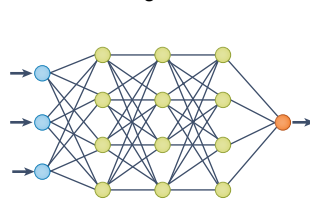
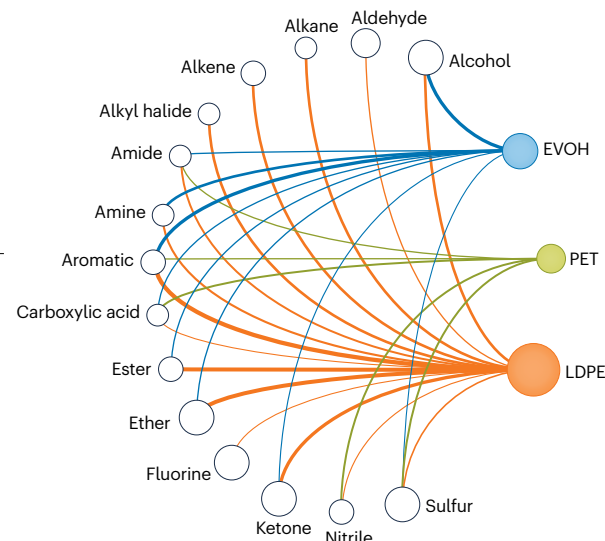
**Modern methods****Molecular modeling****Machine learning****Predicted solubilities**

Fig. 2 | Theories and models for polymer solubility prediction. Four methods to predict polymer solubilities and selective solvents are summarized in the illustrations on the left. The schematic on the right shows different types of solvent able to dissolve example polymers, where the sizes of the circles indicate the number of associated solvents and the widths of the lines indicate solubility

trends⁴⁵. Figure adapted with permission from: left-hand graphic for ‘Modern methods’, ref. 45, Royal Society of Chemistry; right-hand graphic for ‘Predicted solubilities’, ref. 127, Elsevier. Right-hand graphic for ‘Modern methods’ reproduced with permission from ref. 128, Royal Society of Chemistry.

properties) of the recovered resin can be assessed across various categories such as Fourier transform infrared spectra, melt flow index, crystallinity and thermomechanical properties (for example, tensile strength and thermal stability). Studies have shown that the crystallinity³⁸ and mechanical properties³⁹ of solvent-extracted polymers are affected by the process. Limited studies on the purity of solvent-recovered polymers have been carried out. Recently, Deng et al. used dimethyl isosorbide to extract PET from a PET–PS–PVC–PE–PLA blend. No characteristic peaks associated with additives were observed in the ¹H NMR spectra of the recovered PET, showing the high purity of the recovered polymer⁴⁰. Property changes of the recovered polymers can be attributed to the minor degradation of the polymer chains during the dissolution/precipitation stage, the presence of mixtures of polymer architectures in the input material⁴¹ or the removal/degradation of some additives during the process⁴². The recovered polymers can be

further processed to be comparable to virgin resins; however, further investigation into the properties of recovered polymers is necessary³⁹. For example, additives should be reintroduced into the resin, which is not currently addressed in the literature.

Thermodynamics of polymer solubility

Understanding solvent–polymer thermodynamic properties allows for the selection of a suitable solvent and operation conditions for dissolution and precipitation. While some solvents and antisolvents have been identified and tabulated for common polymers⁹, identifying solvents that enable the selective dissolution of a single polymer component from a multicomponent mixture of plastic waste remains a challenge. Figure 2 summarizes four broad categories of theories and computational models that can efficiently guide solvent selection by predicting polymer solubility.

BOX 1

Fundamentals of polymer dissolution

Classical solution theories for polymers focus on the thermodynamics of polymer–solvent mixing, which is governed by the Gibbs free energy of mixing $\Delta G_{\text{mix}} = \Delta H_{\text{mix}} - T\Delta S_{\text{mix}}$. A polymer and a solvent will mix if $\Delta G_{\text{mix}} < 0$. Flory–Huggins theory⁴⁷ is the most prominent solution theory and uses a mean-field lattice model to derive equation (1):

$$\Delta G_{\text{mix}} = RT[n_1 \ln \phi_1 + n_2 \ln \phi_2 + n_1 \phi_2 \chi_{12}] \quad (1)$$

In equation (1), R is the ideal gas constant, T is the temperature, component 1 is the solvent, component 2 is the polymer, n_i is the number of moles of component i and ϕ_i is the volume fraction of component i . The first two terms account for the combinatorial entropy contribution to ΔS_{mix} , while the third term uses the parameter χ_{12} to account for polymer–solvent interactions that contribute to ΔH_{mix} . Solubility parameters are another powerful tool to predict polymer dissolution. Most solubility parameter systems adopt the ‘like dissolves like’ concept and postulate that materials with similar parameter values will be miscible⁵⁶. Notable examples include the Hildebrand parameter and HSPs⁶⁰. The Hildebrand solubility parameter is derived from a material’s cohesive energy density and has reasonable predictive accuracy for non-polar systems, but is limited when there are strong polar or specific interactions such as hydrogen bonds. HSPs extend this concept to three parameters that assess the chemical similarity between a polymer and a solvent on the basis of dispersion (δ_D), polar (δ_P) and hydrogen-bonding (δ_H) interactions. Polymer–solvent interactions are quantified by their virtual distance R_a as shown in equation (2).

$$R_a^2 = 4\left(\delta_D^{\text{solvent}} - \delta_D^{\text{polymer}}\right)^2 + \left(\delta_P^{\text{solvent}} - \delta_P^{\text{polymer}}\right)^2 + \left(\delta_H^{\text{solvent}} - \delta_H^{\text{polymer}}\right)^2. \quad (2)$$

Each polymer has an interaction radius R_0 ; solvents with $R_a < R_0$ are predicted to dissolve the polymer.

Polymer solution theories and solubility parameters have laid the foundation for polymer dissolution behaviors and solubility predictions (Box 1). Phase diagrams based on Flory–Huggins theory have provided physical insight into the key contributions to the free energy associated with polymer–solvent mixing, ΔG_{mix} , at different temperatures and compositions. However, Flory–Huggins theory is a highly simplified physical model and cannot capture all features of a polymer solution, such as the volume change of mixing, the energetically favorable arrangements of molecule segments, or the composition dependence of the interaction parameter⁴³. Hansen solubility parameters (HSPs) are more frequently used to guide solvent selection. HSPs have been tabulated for a wide range of polymers and solvents and are a benchmark method for screening solvents to dissolve polymers⁴⁴. While parameter systems such as HSPs have many successful applications, they only qualitatively predict whether a polymer should dissolve in a target solvent, and generally are less accurate at temperatures other than RT.

Recent studies have applied modern molecular models and computational methods to evaluate polymer solubility in various solvent systems. For example, Zhou et al.⁴⁵ developed a computational approach to predict temperature-dependent polymer solubilities

using molecular-scale models. This approach employs molecular dynamics simulations, conformational sampling techniques and the conductor-like screening model for real solvents (COSMO-RS)⁴⁶. COSMO-RS is a statistical thermodynamics model that can perform solid–liquid and liquid–liquid equilibrium calculations to calculate polymer solubilities. This approach can rapidly predict quantitative polymer solubilities at different temperatures and in both single- and multicomponent solvent systems. Results using this approach have been tabulated as part of a computational polymer solubility database that includes over 10 polymers and 1,000 solvents at multiple temperatures. The database has paved the way to new studies that integrate solvent screening, TEA, LCA and process optimization^{35,43,47} and has successfully been used to guide the design of multistep processes for polymer dissolution and separation in complex mixtures³⁷. Following a similar approach to identify new green solvents, Soyemi and Szilvási⁴⁸ established a solvent library that contains over 9,000 glycerol derivatives and predicted their physicochemical properties using quantum chemical methods, COSMO-RS and machine learning models. The predicted properties included densities, dipole moments, water solubilities, octanol–water partition coefficients, melting points and HSP values. Potential solvent candidates and dissolution strategies for six common polymers were identified on the basis of the predicted data.

Most recently, machine learning methods have been developed to predict properties relevant to polymer dissolution. Chandrasekaran et al.⁴⁹ introduced a solvent/antisolvent classification model using a deep neural network, which Aoki et al.⁵⁰ extended to predict temperature-dependent Flory–Huggins interaction parameters. Ethier et al.⁵¹ combined Flory–Huggins theory with neural networks to improve model interpretability and accuracy. Sanchez-Lengeling et al.⁵² introduced the Gaussian process HSP, a Bayesian method for HSP determination. Kurotani et al.⁵³ used an in-phase deep neural network to predict solubility parameters. Kern et al.⁵⁴ proposed fingerprints of polymers and solvents for room-temperature solubility prediction. These studies largely focus on qualitatively classifying solvents and antisolvents for target polymers or predicting solubility parameters, but a research gap remains in developing machine learning models to quantitatively predict polymer solubilities. Training these models also requires large-scale and high-quality data on polymer solubility, requiring the development of standardized, high-throughput experimental solubility measurement approaches⁵⁵.

This previous discussion has largely focused on polymer solubility at ambient pressure. Another important factor that influences both the enthalpy of mixing (ΔH_{mix}) and the entropy of mixing (ΔS_{mix}) is solvent density^{56,57}. In a homogeneous solution, the contribution of pairwise interactions to ΔH_{mix} is roughly proportional to the density, which approximately determines the number of interactions per volume. Increasing solvent density for a system with favorable polymer–solvent interactions can thus increase polymer solubility⁵⁷. However, if there are large differences in the densities of the solvent and polymer, the corresponding change in volume upon mixing also contributes to ΔS_{mix} (ref. 56). Unlike the combinatorial entropy, which always promotes mixing, the non-combinatorial contribution to ΔS_{mix} due to this density difference disfavors mixing by reducing the entropy of the lower-density (typically solvent) component. This behavior leads to phase separation at temperatures above the lower critical solution temperature⁵⁶; at high temperatures, the solvent density decreases to a much greater degree than the polymer density and the non-combinatorial entropy is highly unfavorable. The impact of solvent density on polymer solubility has motivated interest in supercritical fluids as solvents for polymer dissolution⁵⁸. Unlike typical solvents, which are nearly incompressible, the density of a supercritical fluid can be varied from gas-like to liquid-like by manipulating the pressure. Increasing the pressure of a supercritical fluid such that its density approaches that of a target polymer will drive dissolution by reducing the unfavorable non-combinatorial

entropy⁵⁷. Pressure variations can also lead to differences in solubility for polymers with similar chemical properties but distinct densities (for example, low-density PE (LDPE) versus high-density PE (HDPE))⁵⁹. The addition of pressure as a 'knob' to tune the properties of supercritical fluids has motivated the use of solvents with low critical temperatures and pressures, such as carbon dioxide, ethylene propane and butane, to dissolve and purify polymers; one such example is the PureCycle process (discussed below).

Together, the exciting developments in physics-based and data-driven predictions of polymer solubility support the ability of modern computational methods to screen large sets of solvents, including new solvents for which experimental data may not be available. Ongoing challenges include treating variations in polymer solubility associated with molecular weight, crystallinity⁶⁰ and the presence of small-molecule components, such as common additives, pigments or contaminants in plastic waste⁶¹. Predictive methods for solubility are also less established for supercritical fluids than for typical solvents. Current methods rely on equation-of-state models, such as the perturbed chain–statistical associating fluid theory equation of state⁶², and some recent methods have developed machine learning methods or extended COSMO-RS models appropriate for predicting solubility in supercritical carbon dioxide⁶³, but more generalizable methods are needed. Other challenges include predicting solution properties other than solubility, such as viscosity, that will influence the design of solvent-mediated recycling methods²⁸. For example, thermodynamic models to understand solubility are also useful for understanding the thermodynamics of precipitation as a method to recover dissolved polymer resins from solution. However, polymer crystallization during precipitation depends upon the thermodynamics of nucleation, which remains challenging to predict⁶⁴. The kinetics of polymer crystallization during precipitation can likewise greatly affect the properties of the recycled materials. For example, crystallization temperature, cooling rate and the application of high shear strain are typical factors to control the crystal structure and morphology of polymers⁶⁴. Therefore, while polymer solubility is a critical factor in solvent selection, both thermodynamic and kinetic factors need to be considered in process design. Because the thermodynamic properties of solvents, polymers and impurities strongly influence the operating conditions of the recycling process and its overall economic and environmental performance, another important challenge is the co-design of solvents and of the recycling process (particularly when diverse polymers and impurities are present in the input waste feedstock)⁶⁵.

Commercial activities in dissolution-based recycling

Summary of current dissolution-based plastic recycling plants

Table 2 summarizes several pilot- and industrial-scale dissolution-based recycling technologies that have been reported, including their nameplate capacities, locations, required feedstocks, products and the solvents and dissolution conditions. The current operational status of each of these facilities is not clear.

CreaSolv

CreaSolv is a plastic recycling process developed by the Fraunhofer Institute for Process Engineering and Packaging IVV^{66,67}. According to their official website, the process involves solvent selection, targeted polymer dissolution, removal of contaminants from the solution, precipitation of the polymer, drying and final product processing⁶⁸. The process flow diagram is summarized in Fig. 1a. CreaSolv did not disclose the specific types of solvent they use for their process, apart from defining HSP domains in their patent series⁶⁹. As shown in Table 1, CreaSolv has three small industrial facilities or pilot plants that produce (1) PE from sachets, (2) PS from foam waste and (3) PVC from floor covering.

APK Newcycling

APK Newcycling operates an 8,000-ton-per-year plant in Merseburg, Germany, that has been in operation since June 2018⁷⁰. Their process extracts PE and PAs from post-industrial multilayer plastic wastes. According to one of the patents published in 2022, the process involves the shredding and categorization of feedstock, washing and separation through density differences, dissolution of targeted polymer, solid and liquid separation using a centrifuge, cross-flow filtration with a membrane for purification, flash evaporation and devolatilization for polymer extraction, and extrusion to produce final product (see Fig. 1b)⁷¹. The company has partnered with Coperion to incorporate ZSK twin-screw extruder technology into its process, ensuring efficient polymer devolatilization and extrusion⁷².

Polystyvert

Polystyvert has demonstrated a 600-ton-per-year plant since August 2018 in Montreal, Canada, focusing on the recovery of PS⁷³. According to a patent published in 2023, the process uses PS waste, such as foam packaging, as feedstock⁷⁴. The material is dissolved in *p*-cymene, followed by filtration and precipitation using a non-polar antisolvent such as heptane or hexane. The precipitated PS undergoes several washing and separation steps to remove impurities, after which it is dried and extruded. The process flow diagram can be found in Fig. 1c. One of the key innovations lies in its solubilization stage, where the solvent is also able to dissolve various non-polar additives such as hexabromocyclododecane and silicone oils. The dissolution equipment utilizes a metallic grid to help push the expanded PS into the solvent, which further reduces the dissolution time⁷⁵. To advance the commercial scale-up of this technology, Polystyvert partnered with INEOS Styrolution in 2021⁷⁴.

PureCycle

PureCycle Technologies is commercializing a proprietary process developed by Procter and Gamble that uses supercritical butane to recycle PP plastic waste into high-quality resin. The company worked with KraussMaffei, a company that specializes in degassing extruders, to upscale its process⁷⁶. A simplified process flow diagram is shown in Fig. 1d ref. 32. The process begins with preprocessing feedstock. According to a report published by Leidos, the preprocessing for high-purity PP wastes involves feedstock de-baling, shredding, metal removal, granulating and agglomeration⁷⁷. After preprocessing, the feedstock is fed into an extruder to melt the material and remove solid impurities. The melted PP is then fed into a light extraction column, to extract and remove light, soluble impurities. The solvent is then directed to a flash tank for separation. The remaining PP–solvent mixture enters a mixing column to fully dissolve the PP. Subsequently, the mixture moves into a settler, and the heavy phase is conveyed out of the settler for scrap storage. Additionally, the PP–solvent mixture from the settler passes through candle filters to remove insoluble solid impurities, after which adsorption columns are utilized to remove remaining impurities. The PP solution is then reheated to further remove the solvent. The PP undergoes further pelletization for storage or reprocessing as needed.

STRAP

STRAP is building a 200-ton-per-year pilot system, which can process mixed plastic wastes including multilayer plastic to extract high-quality resin, in the United States³⁰. Figure 1e shows the simplified process flow diagram of STRAP. After the feedstock is preprocessed, it is selectively dissolved and undergoes hot filtration. The undissolved material is discharged and dried to recover solvent, while the solution undergoes hot adsorption to remove small-sized impurities. The solution is then cooled to allow precipitation, after which the precipitated polymer is separated and dried for resin production and the solvent is purified to be reused in the process. Due to the solvent traces trapped

Table 3 | Summary of key technical, economic and environmental variables for different dissolution-based processes

Processes (precipitation method)	Feedstock	Recovered polymers	Plant capacity (ton _{yr} ⁻¹)	IRR (%)	Capital cost (million US\$)	Solvents (solvent/antisolvent)	Solvent recycling rate	Climate change impact (kg CO ₂ kg ⁻¹)	Climate change impact (% relative to virgin)	Target economic and environmental metrics	Key economic and environmental impact drivers	Refs.
Polystyrene—commercial (A)	Post-industrial and post-consumer waste with PS, EPS and HIPS	PS, EPS, HIPS	9,000	—	40	Ethyl benzene, <i>p</i> -cymene/proprietary	Not reported	Not reported	−90% ^a	Capital investment and GHG emissions	Not reported	95
CreaSolv I—commercial, electronic waste (C)	Electronic scraps	HIPS, PP, PLA	3,000	—	Not reported	Proprietary (cycloaliphatic hydrocarbons)	‘High’	Not reported	Not reported	Not reported	Not reported	67
CreaSolv II—commercial, PS Loop (C)	EPS	PS	3,000	—	Not reported	Proprietary	Not reported	Not reported	Not reported	GHG emissions	Not reported	122,123
Purecycle—commercial (C)	Carpet, PCR bags/woven sacks, rigid, biaxially oriented PP	PP	48,534	—	276.51	Butane	Not reported	Not reported	−35% ^a	Capital investment, GHG emissions	Not reported	77,124
APK—commercial (O)	Post-industrial and post-consumer waste	LDPE, nylon	20,000	—	Not reported	Formic acid, methylcyclohexane	Not reported	Not reported	−66% ^a	Not reported	Not reported	70,72
HDPE dissolution (A)	Baled waste with 90 wt% of HDPE	HDPE	54,750	10	26.7	Xylene/hexane	97%	2.40	+85%	MSP, GHG emissions, energy use, land use, toxicity, waste generation, water use and electricity	Solvent recycling rate and process utilities (steam and electricity)	13
LDPE dissolution (A)	Baled waste with 95 wt% of LDPE	LDPE	54,750	10	27.2	Xylene/hexane	97%	2.24	+22%			
PET dissolution (A)	Baled waste with 88 wt% of PET	PET	54,750	10	26.3	Benzyl alcohol/methanol	98%	4.49	+32%			
PP dissolution (A)	Baled waste with 92 wt% of PP	PP	54,750	10	26.7	Xylene/hexane	97%	2.16	−19%			
Antisolvent—PET (A)	Baled waste with 90 wt% of PET	PET	8,400	20	11.2	GVL/water	98%	3.55	+59%	MSP, net present value, capital investment, return on investment, GHG emissions, energy demand	Feedstock cost, solvent recycling rate, process utilities	86
Cooling—PET (C)	Baled waste with 90 wt% of PET	PET	8,400	20	10.3	GVL	98%	1.08	−52%			
Evaporation—PET (O)	Baled waste with 90 wt% of PET	PET	8,400	20	8.5	GVL	98%	1.20	−46%			
STRAP-A (A)	Post-industrial waste of multilayer films with ~90 wt% of PET	PE w/EVA, EVOH, PET	3,000	10	25.6	Toluene, DMSO/acetone, water	>99%	6.73	+150%	MSP, capital investment, GHG emissions, energy use, toxicity, water use	Solvent/polymer ratio, process utilities (steam and electricity)	82

Table 3 (continued) | Summary of key technical, economic and environmental variables for different dissolution-based processes

Processes (precipitation method)	Feedstock	Recovered polymers	Plant capacity (ton _{yr} ⁻¹)	IRR (%)	Capital cost (million US\$)	Solvents (solvent/antisolvent)	Solvent recycling rate	Climate change impact (kg CO ₂ kg ⁻¹)	Climate change impact (% relative to virgin)	Target economic and environmental metrics	Key economic and environmental impact drivers	Refs.
STRAP-B (C)	Post-industrial waste of multilayer films with ~90 wt% of PET	PE, EVA, EVOH, PET	3,000	10	22.4	Toluene, DMSO, water/ acetone	>99%	1.17	-57%			
STRAP-C (O)	Post-industrial waste of multilayer films with ~70 wt% of PET	PE, EVA, EVOH, PET, PET glycol-modified	3,000	10	31.8	DMF, toluene, DMSO, water/-propanol, acetone	>99%	2.07	-21%			
STRAP-ink I (C)	Post-industrial waste of printed multilayer films with ~60 wt% of PE	PE	3,000	10	18.2	Dodecane	>99%	0.96	-60%	MSP, capital investment, GHG emissions	Interest rate, process utilities (steam and electricity), solvent recycling rate	83
STRAP-ink II (C)	Post-industrial waste of printed multilayer films with ~60 wt% of PE	PE, EVOH	3,000	10	26.4	Dodecane, DMSO, water	>99%	1.10	-64%			
STRAP-ink III (C)	Post-industrial waste of printed multilayer films with ~60 wt% of PE	PE, EVOH, PET	3,000	10	37.8	Dodecane, DMSO, water, GVL	>99%	1.48	-51%			
STRAP-PP I (C)	Disposable facemasks with ~80 wt% of PP	Colored PP (light blue)	5,300	10	24.7	Toluene	>99%	0.57	-72%	MSP, capital investment, GHG emissions	Interest rate, solvent/polymer ratio, process utilities (steam and electricity)	91
STRAP-PP II (C)	Disposable facemasks with ~80 wt% of PP	PP	5,300	10	32.2	Toluene, DMAc	>99%	0.87	-58%			
Dissolution—textiles (O)	Elastane/polyester textile waste with 10 wt% of elastane	PET	—	—	Not reported	DMF/tetrahydrofurfuryl alcohol	99.99%	1.19	-59%	GHG emissions	Solid/liquid ratio, solvent recovery rate	92
Solvent-based PLA (O)	Lightweight packaging waste stream with 8% of PLA	PLA	—	—	Not reported	Composition not stated due to non-disclosure	99.99%	0.52	-74%	GHG emissions, fossil resource and ozone depletion, land use, acidification, freshwater, marine and terrestrial eutrophication	Process utilities (steam and electricity)	125

*All climate change values from commercial processes were taken from official company websites or publications. IRR is the discount rate. PCR is post-consumer resin.

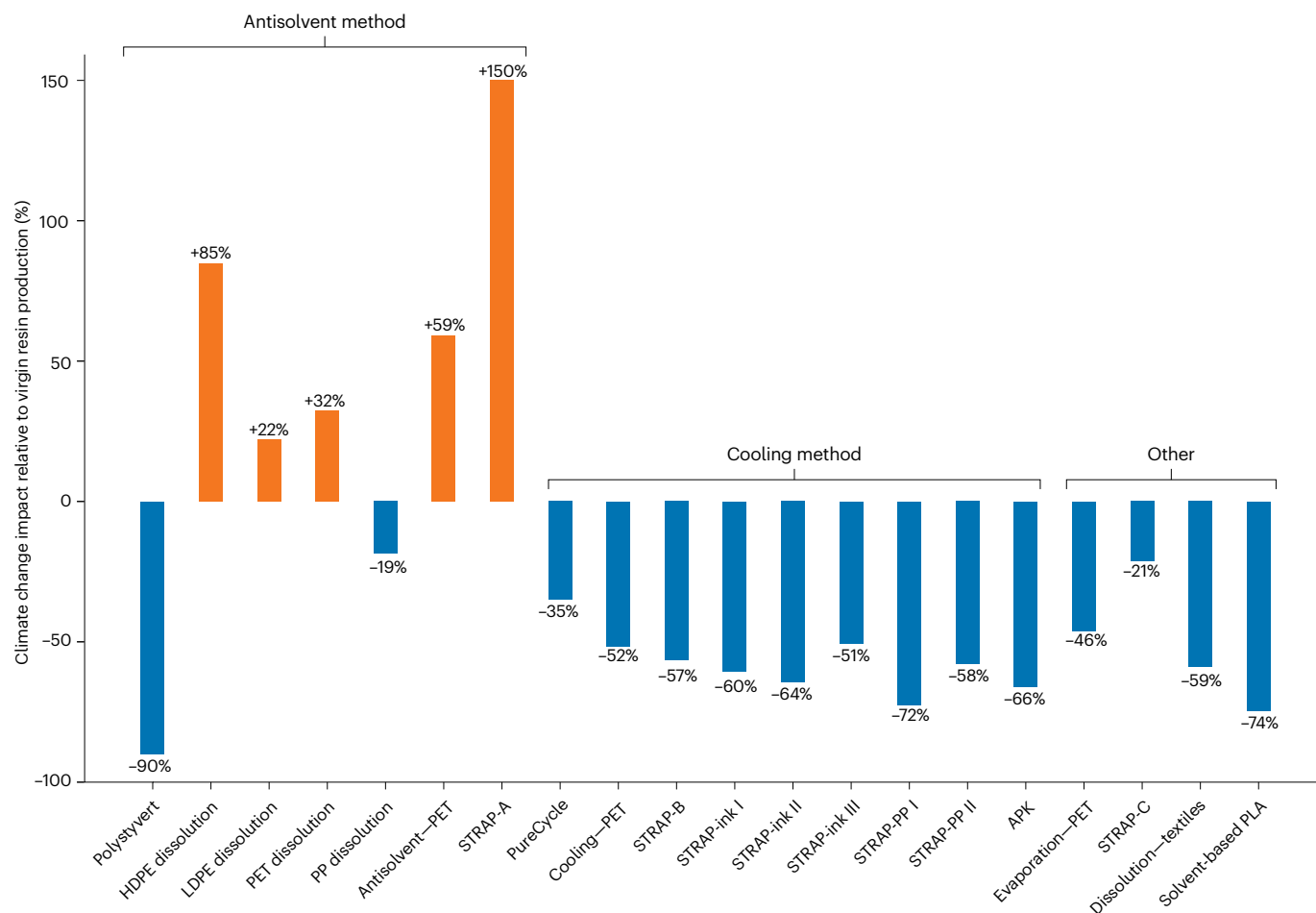


Fig. 3 | Comparison of climate change impact (kg CO₂e per kg resin) for reported dissolution recycling processes and their respective virgin resin counterparts. Dissolution-based processes are compared with virgin resin production. The processes are differentiated by precipitation method:

antisolvent, cooling or other. A positive percentage represents an increase in GHG emissions while a negative percentage represents a decrease in GHG emissions. All referenced values can also be found in Table 3.

after drying of the solids, makeup solvent is required to sustain the continuous process.

Impact of dissolution-based recycling on quality of recycling

Dissolution-based recycling processes have the potential to provide economic and environmental benefits by reducing resource consumption and improving the quality of recycling. Therefore, next to technical quality of the outputs, TEA and LCA studies play a vital role in assessing and guiding the design of these technologies.

Technical quality of recycled materials

In several studies, solvent treatment did not significantly affect the mechanical, rheological or thermal properties of the final polymer^{78,79}. However, in some cases, slight differences in the molecular weight, thermal properties and melt processability of the recovered polymer were observed due to possible degradation or cross-contamination with other components present in the system^{80,81}. Another factor that can influence the polymer properties after recycling is solvent retention, which can change the melt behavior of the polymer by reducing its viscosity and which can affect the final use (for example, in food-grade applications)⁸¹. Additionally, differences in the process design, such as using different dissolution solvents and processing temperatures, can influence the polymer properties²⁵. Property comparisons are important to understand whether dissolution-based

recycling processes can produce high-quality polymers, and additional attributes to characterize include tensile strength, flexural strength, impact strength, membrane performance and polymer solution characterization (density, refractive index, viscosity)²⁵. The materials recovered via dissolution-based processes can have potential applications in packaging films, membranes, glues, paint additives, natural/fiber composites and textiles, depending on the recovered material and its properties. Therefore, dissolution-based processes can help produce high-quality products from diverse feedstocks that are hard to recycle (Table 2) and be used as a separation technology for multicomponent complex waste. Some limitations include separating mixtures with specific polymers that are challenging to dissolve (for example, PET and nylons). Previous approaches have overcome this issue by leaving the challenging component (for example, PET) at the end of the separation sequence and recovering it as a residual solid without being dissolved²⁵. Two other challenges are the separation of different grades of the same polymer and the removal of impurities such as dyes, which impact the color and final use of the recycled materials. Often, in the case of these more challenging separations, costly process steps such as elevated temperatures, pressures or added separation steps are required to achieve high-quality resins^{13,82–85}. More research is needed to outline the trade-offs between product quality and process costs. Demonstrating the food safety of polymers recovered from dissolution recycling also needs further research. Nevertheless, at least on a theoretical basis,

dissolution-based recycling has the potential to overcome these challenges and produce virgin-like resins.

Economic analysis

Academic and industry reports have shown that the economic feasibility of dissolution-based processes depends strongly on the precipitation method, feedstock type, recovered polymers, economies of scale (plant capacity) and solvents (type and recycling rate)^{13,82,84,85}. Table 3 summarizes key variables for different processes in various stages of development. The processes were differentiated by name and precipitation method including antisolvent (A), cooling (C) or other (O), which comprise a combination of methods. These data are primarily based on laboratory studies, rather than on commercial data, which are difficult to obtain. While estimates of economic and environmental impacts from laboratory data can have limited accuracy, comparing different processes and studies helps identify the main impact drivers and general trends.

Table 3 presents the capital costs for different processes at different plant capacities. These costs are heavily dependent on the required type of equipment. A trend of similar costs in processes with equal capacities and assumptions can be seen (for example, antisolvent–PET and cooling–PET)⁸⁶. However, the technology that uses antisolvent as a precipitation technique results in higher capital costs. The same trend can be observed comparing STRAP-A, which uses antisolvent, and STRAP-B, which uses cooling⁸⁷. These studies have reported that the need for distillation steps to separate the solvent/antisolvent mixture and the required process utilities lead to higher capital and operating costs. In addition, it has been reported that the solvent/polymer ratio and non-feedstock inputs (mainly solvent) are large contributors to process economics⁸⁷. Therefore, high polymer concentrations with high solvent recycling rates are typically necessary to achieve profitability.

Feedstock type and composition are also key economic drivers; some processes may have to purchase feedstock with prices depending on the quality and availability, while others may receive a tipping fee for accepting waste⁸⁶. To date, there is no clear trend between feedstock quality and cost. Some companies, such as PureCycle, have invested in sorting technologies to source ideal feedstocks, but this does not mean that high-quality low-cost feedstocks such as post-industrial waste are not available⁸⁸.

Feedstock composition plays a key role because prices of recovered materials can vary (for instance, HDPE may sell for US\$1 kg⁻¹ and EVOH may sell for US\$8 kg⁻¹)^{89,90}. These differences can drive process profitability and design (number of solvent washes to recover all polymers or only the most valuable components). It is also important to highlight that certain polymers require higher energy inputs to dissolve (due to unfavorable thermodynamics), and contaminated feedstocks may need prewashing and other purification steps⁹¹. Solvent choice impacts energy requirements for solvent removal and affects product use (for example, only some solvents may be suitable for food contact applications).

Many studies report that dissolution-based recycling can produce high-quality recycled resins at prices competitive with those of virgin polymers (using the minimum selling price (MSP) as a metric)^{86,87,91}. The recovered resins are typically cleaner than mechanically recycled resins and more valuable than the oils or monomers produced by chemical recycling, offering a competitive advantage over other recycling methods⁸⁴. This suggests that dissolution-based recycling can be profitable despite initial capital expenditures, if there is sufficient volume to achieve favorable economies of scale. The larger commercial processes aim to recycle at 20,000–50,000 metric tons per year^{70,77}; this is the current best indicator for necessary scale, but operational data are lacking. It is important to note the significant uncertainty in current academic and industrial preliminary estimations, which require more detailed studies in pilot systems.

Feedstock selection can have a large impact on capital costs for dissolution recycling plants and should be carefully considered during

the commercialization of this new technology. Cleaner feedstocks such as post-industrial multilayer waste require fewer preprocessing and contaminant removal steps, which can considerably lower capital expenses. Similarly, colorless plastic waste may require fewer treatment steps and save costs. If available, plastic waste containing only two polymers presents an opportunity for plants to recover two products with a single solvent, improving the plant's economic efficiency. If possible, early commercial solvent-based recycling plants should focus on 'ideal' feedstocks first, moving to more complicated waste streams as the technology is better tested and understood.

Environmental impact analysis

The environmental impacts of dissolution-based processes have been quantified by various LCA methodologies. Most studies only report the climate change impact or equivalent greenhouse gas (GHG) emissions of the process^{82,92}. However, other environmental metrics, such as resource use, water use, land use, toxicity, material circularity and others, can also be relevant, as the single metric of climate change is likely insufficient in determining all environmental impacts of a process⁹. For instance, modern engineered landfills are the most used plastic waste management process, and generate minimal GHG emissions, although their environmental impacts related to soil and groundwater pollution are substantial⁹. Additionally, current LCA approaches have not standardized the selection of system boundaries⁹³. This lack of consistency hinders the comparison between approaches and can result in inconsistent or misleading conclusions⁹⁴.

Table 3 provides a summary of the climate change impact and key drivers reported by different references. Most studies use a cradle-to-gate approach, and depending on the type of waste (post-industrial or post-consumer) some approaches include pretreatment^{9,86}. A common environmental impact driver in the processes that use antisolvent is steam, which is required to separate the solvent–antisolvent mixture via distillation. The climate change impacts reported with the virgin production alternative are compared in Fig. 3. It is important to note that these studies may use different assumptions, and reported values may become outdated with updates to technology or LCA methodology. It can be found that, in general, the processes with a higher environmental impact than the virgin alternative use antisolvent as a precipitation method. The specific increase in GHG emissions appears in Fig. 3 as a positive percentage highlighted in red. This impact is a direct result of the energy required to separate the solvent and antisolvent and to recycle them. On the other hand, the processes that use cooling or other methods have a lower climate change impact than virgin polymer production because of lower energy requirements. These methods also include solvent recovery and recycling, but the energy demand is lower because there are no solvent mixtures (see the above discussion for more details on solvent recovery). These results again highlight the need to identify solvents that lead to favorable process designs and operating conditions. The lowest emissions were reported by the company Polystyvert, followed by Solvent-based PLA and STRAP-PPI. Some important characteristics of the Polystyvert–commercial process are the use of low operating temperatures, an essential oil as solvent and the removal of the flame retardant hexabromocyclododecane⁹⁵. However, there are no publications available that outline their LCA assumptions and methodology. The other processes with low reported impacts recover only one component and do not use antisolvents for the precipitation. Previous studies reported the solid/liquid ratio and process energy consumption as the main contributors to GHG impacts and identified that high solvent recycling rates are essential to reduce emissions⁹. Furthermore, the environmental benefits of the different dissolution-based processes are a function of the material that is treated and the number of recovered polymers^{42,48}. All commercial processes report a reduction in climate change impact and use multicomponent waste as feedstock^{67,77}.

As shown in Fig. 3, reported dissolution processes without anti-solvents have a GHG impact of approximately 1–2 kg CO₂-equivalent per kg plastic produced, roughly 50% lower than virgin resin production. This is also lower than typical reported emissions for other advanced recycling methods such as glycolysis or pyrolysis^{96–98}. In general, other impact categories such as energy consumption follow a similar trend^{82,84}. However, comparison across analysis methods is challenging, as factors such as energy mix and waste credits can have a major impact, further highlighting the need for consistent LCA approaches.

Almost all chemical processes have economic and environmental trade-offs; however, the evaluated dissolution-based recycling processes agree on common drivers for cost and GHG emissions. For instance, the need for high solvent recycling rates and cooling as a precipitation method has been consistently shown to lead to economic and environmental benefits (mainly focusing on GHG emissions). Of course, solvent consumption is also related to the polymer concentration used in the system. While reducing solvent use lowers environmental impact, it also increases solution viscosity, and maintaining a viable working viscosity takes precedence over minimizing solvent consumption. There is still significant uncertainty in current TEA and LCA approaches; therefore, these conclusions may change as dissolution-based technologies evolve and other environmental metrics are evaluated.

Challenges for commercialization

It is unknown if commercial dissolution-based technologies are operating at full capacity. Several of the facilities have been reported to have operational challenges, which are to be expected for pioneer process plants. For instance, a PVC recycling plant in Europe that was designed to recycle 10,000 ton yr⁻¹ of PVC ceased to operate in 2018⁶⁸. Publicly available information indicated that this closure was partly due to the new legislation that identified the phthalates used in their process being hazardous⁹⁹. Another 49,000-ton-per-year plant built in North America is currently operating at lower than its design capacity¹⁰⁰. It is thus clear that dissolution-based recycling still has several challenges, often related to chemical engineering. First, a solvent must be chosen that only dissolves the targeted polymer and only minimally dissolves any of the other polymers, the pigments or any additives. The undissolved solids are then removed by filtration or centrifugation from a viscous polymer solution without clogging. Depending on the polymer, the dissolution temperature can also be relatively high (for example, >130 °C for polymers such as PP and PET)^{45,101}, which means that the processes, equipment, tubing and so on need to withstand high-temperature solvent environments, while every cold spot can induce polymer precipitation and clogging. After the solid–liquid separation, the dissolved impurities should be removed, for example by an adsorption bed. However, adsorption from mostly apolar solvents of additives is not well studied⁶¹, compared with, for example, adsorption in wastewater treatment, in which the process starts from a polar matrix (water). Afterwards, the polymer must be precipitated from the solution, which could cause gel formation if not well operated³⁵. Avoiding gel formation and forming good polymer precipitates is a challenge in dissolution-based recycling. Limiting solvent entrapped in the precipitates reduces the need to remove high amounts of solvent from the resulting gel. The polymer has to be dried to the appropriate level (typically in the ppm range). Dissolution-based recycling thus requires a completely different plant design compared with traditional mechanical recycling plants, as dissolution plants could deal with volatile/flammable solvents¹⁰². Therefore, correct safety/hazard engineering controls need to be built into any system. Further details regarding the chemical engineering challenges associated with the commercialization of dissolution-based plastic recycling are summarized in Supplementary Table 1.

Key recommendations for the future

Dissolution-based plastic recycling technologies have shown considerable potential in producing high-quality resins from diverse plastic waste streams. The thermodynamics of pure polymer–solvent interactions are well documented, and TEA and LCA data indicate that these methods are potentially cost-effective and environmentally sustainable, with several companies now developing industrial plants.

However, several challenges must be addressed to advance this technology. (1) Preprocessing and sorting: enhanced techniques for sorting plastic waste are crucial to optimize the recycling process. (2) Additives in plastics: research is needed to better understand what happens to additives during dissolution and how to remove additives if desired. (3) Polymer solution properties: improved predictions and management of polymer solution properties, such as viscosity, are essential for optimizing the dissolution and precipitation stage of the process. (4) Solvent reuse and impurity accumulation: investigating the impact of impurity build-up in reused solvents and develop methods to mitigate their effects is vital. (5) Recovered-resin properties: further study is required to understand and enhance the properties of recovered resins for various applications.

Some of the above challenges can be addressed in part by policy. For example, improved sorting practices can be required, or dangerous additives can be outlawed. However, many of the challenges related to dissolution recycling are chemical engineering challenges. By addressing these challenges through targeted research and development, dissolution-based plastic recycling technologies can be optimized and integrated into the broader plastic recycling landscape, providing a sustainable solution for managing plastic waste, and making them a key solution for sustainable plastic waste management.

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Author contributions

G.W.H. supervised the project. Z.X. conceived of the Review, coordinated authors, and combined and unified author contributions. All authors contributed to discussing, commenting on, writing and revising the manuscript.

Competing interests

The following authors are inventors on at least one of the following patent applications related to this work: Z.X., K.S.-R., C.G., P.Z., R.C.V.L., E.B.-Z. and G.W.H. Patent applicant, Wisconsin Alumni Research Foundation; inventors, G.W.H., K.S.-R., R.C.V.L., T. Walker, P.Z.; application number, 17/989960; status of application, pending;

specific aspect of manuscript covered in patent application, dissolution recycling methodology and computational methods for selecting recycling conditions. Patent applicants, Wisconsin Alumni Research Foundation and Michigan Technological University; inventors, G.W.H., Z.X., K.S.-R., E.B.-Z., S. Kolapkar, V. Ierulli; application number, 17/892573; status of application, pending; specific aspect of manuscript covered in patent application, examples of process steps, methodology and equipment. Patent applicants, Wisconsin Alumni Research Foundation and Michigan Technological University; inventors, G.W.H., T. Yan, C.G., K.S.-R., R.C.V.L., E.B.-Z., F. Long, P.Z.; application number, 18/665208; status of application, pending; specific aspect of manuscript covered in patent application, methods to remove solvent and color bodies from plastic, methods to remove color bodies from solvent. All other authors declare no competing interests.

Additional information

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